DS 7330
File Organization and Database Management
Mini Project 7

Name:

This is a mini project for DS 7330, File Organization and Database Management. For this assignment, turn in a single pdf file containing all of your answers. The file should be named ¡yourLastName¿MiniProject-Number.pdf. For example, the file name for my mini project 1 would be 'RafiqiMiniProject-1.pdf'. Insert your answer pages into this file with the answer for Question 1 inserted immediately after Question 1 and before Question 2, the answer for Question 2 inserted immediately after Question 2, and the answer for Question 3 inserted immediately after Question 3. You may insert a front page containing your name and date if you do not wish to or cannot electronically add that information to the first page of this homework sheet.

Collaboration is expected and encouraged; however, each student must hand in their own homework assignment. To the greatest extent possible, answers should not be copied but, instead, should be written in your own words. Copying answers from anywhere is plagiarism, this includes copying text directly from the textbook. Do not copy answers. Always use your own words and your own code. Directly under each question list all persons with whom you collaborated and list all resources used in arriving at your answer. Resources include but are not limited to the textbook used for this course, papers read on the topic, and Google search results. Don't forget to place your name on the first page of the pdf document.

Objective:

In this assignment, you'll explore how to use **partitioning** techniques in MySQL to optimize the unified database containing employee and customer information. This exercise will involve creating partitions, setting up clustered indexes, and understanding their impact on database performance. Use the **InClassDB** to answer these questions.

Question 1: Partitioning the Orders Table by Year

Task: Partition the `Orders` table by **year** using the `OrderDate` column. Write a query to fetch all orders from a specific year and explain the impact of partitioning on query performance.

Instructions:

1. Create a partitioned version of the `Orders` table using **range partitioning** based on the year in `OrderDate` (years e.g., 2022, 2023, 2024)
2. Write a SQL query that retrieves all orders from the year 2023.
3. Explain how partitioning the table by year affects query performance and data management.

Note: If there is no data for those ranges, please add a few records so that you have data for all date ranges within your partitions.

Submission: Submit the partitioning SQL script, the query, and an explanation of the performance benefits.

Question 2: Partitioning the Products Table by Product Category

Task: Partition the `Products` table by **Product Category** using a **list partition** strategy. Create separate partitions for product categories such as 'Electronics', 'Clothing', and 'Food'. Write a query to fetch all products from one category and discuss how partitioning impacts data retrieval.

Instructions:

1. Create a partitioned version of the `Products` table using **list partitioning** based on the `ProductCategory`.
2. Write a SQL query that retrieves all products belonging to the 'Electronics' category.
3. Explain the benefits and limitations of partitioning the table based on categories.

Submission: Provide the partitioning SQL script, the query, and a brief discussion on the effectiveness of partitioning by product category.

NOTE: It may be difficult to notice performance improvements due to the data size, in that case please provide your understanding as to why partitioning would/could improve performance.

Resources: ____________

Collaborators ______________